Prayas JEE 2.0 (2024)

Quadratic Equations

DPP-06

1. If one root of the equation $x^2 + px + q = 0$ is the square of the other, then **[IIT-JEE-2004]**
(1) $p^3 + q^2 - q(3p + 1) = 0$
(2) $p^3 + q^2 + q(1 + 3p) = 0$
(3) $p^3 + q^2 + q(3p - 1) = 0$
(4) $p^3 + q^2 + q(1 - 3p) = 0$

2. If α and β are the roots of the quadratic equation, $x^2 + x\sin\theta - 2\sin\theta = 0, \theta \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\alpha^{12} + \beta^{12}}{\left(\alpha^{-12} + \beta^{-12}\right)(\alpha - \beta)^{24}}$ is equal to: **[JEE Mains-2019]**
(1) $\frac{2^6}{(\sin\theta + 8)^{12}}$
(2) $\frac{2^{12}}{(\sin\theta - 8)^6}$
(3) $\frac{2^{12}}{(\sin\theta - 4)^{12}}$
(4) $\frac{2^{12}}{(\sin\theta + 8)^{12}}$

3. Find the range of values of a, such that $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative.

4. Find the least value of n such that $(n - 2)x^2 + 8x + n + 4 > 0$, $\forall\ x \in R$, where $n \in N$.

5. If $x \in R$, then find range of $\frac{x+2}{2x^2 + 3x + 6}$.

6. The smallest value of $x^2 - 3x + 3$ in the interval $(-3, 3/2)$ is
(1) 3/4
(2) 5
(3) –15
(4) –20

7. Consider a rational function $f(x) = \frac{x^2 - 3x - 4}{x^2 - 3x + 4}$.
The sum of integers in the range of $f(x)$, is:
(1) –5
(2) –6
(3) –9
(4) –10

8. Given that, for all $x \in R$, the expression $\frac{x^2 - 2x + 4}{x^2 + 2x + 4}$ lies between $\frac{1}{3}$ and 3, the values between which the expression $\frac{9.3^{2x} + 6.3^x + 4}{9.3^{2x} - 6.3^x + 4}$ lies, are
(1) – 3 and 1
(2) 1 and 3
(3) – 1 and 1
(4) 0 and 2

9. If the roots of equation $x^2 + (p + 1)x + 2q - q^2 + 3 = 0$ are real and unequal $\forall\ p \in R$, then the minimum integral value of $(q^2 - 2q)$ is
(1) 3
(2) 4
(3) 5
(4) 6

10. If $a_1, a_2, a_3, a_4, \ldots, a_{n-1}, a_n$ are distinct non-zero real numbers such that
$$\left(a_1^2 + a_2^2 + a_3^2 + \ldots + a_{n-1}^2\right)x^2 + 2\left(a_1a_2 + a_2a_3 + a_3a_4 + \ldots + a_{n-1}a_n\right)x + \left(a_2^2 + a_3^2 + a_4^2 + \ldots + a_n^2\right) \le 0,$$
then $a_1, a_2, a_3, a_4, \ldots, a_{n-1}, a_n$ are in **[Challenger]**
(1) A.P.
(2) G.P.
(3) H.P.
(4) A.G.P.

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (4)
2. (4)
3. $a \in \left(-\infty, -\frac{1}{2}\right)$
4. (5)
5. $\left[-\frac{1}{13}, \frac{1}{3}\right]$
6. (1)
7. (2)
8. (2)
9. (2)
10. (2)